

Minjie Xu

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Address: Wuhan Textile University, 430200 Wuhan, China

Profile

MSc candidate in Textile Science and Engineering at Wuhan Textile University, working at the State Key Laboratory of New Textile Materials and Advanced Processing Technologies, focusing on biomass-based materials and the development of novel functional textiles. **Research interests** include collagen-based biomaterials, electrospun nanofibers, high-performance fiber materials, multifunctional textile materials, and wearable sensing systems. Experienced in the fabrication and characterization of functional materials, with skills in electrospinning, nanomaterial integration, and material performance evaluation.

Personal Attribute

Open-minded and curious researcher with strong logical thinking, problem-solving ability, and teamwork skills.

Publication

Xu M, et al. Tannic Acid-Mediated Multifunctional Collagen/Cellulose Nanofiber Composite Aerogels for Sustained Drug Release, Antibacterial Properties, and Hemostasis [J]. ACS Applied Polymer Materials, 2025, 7(15): 9750-63.

Yue C, **Xu M**, et al. pH-responsive release features of collagen/carboxylated cellulose nanofiber composite aerogels through the incorporation of cyclodextrin/5-fluorouracil inclusion complexes[J]. European Polymer Journal, 2024, 207.

Education

(Sep 2023 - Present)

Master of Materials and Chemical Engineering (GPA: 3.38/4.00) :

Wuhan Textile University, Wuhan, China.

Specialization: Preparation and Characterization of Collagen-Based Composite Materials.

Main courses: Fiber Physics and Chemistry, Textile Materials Science, Material testing and characterization techniques.

(Sep 2021 - Sep 2023)

Bachelor of Engineering in Light Chemical Engineering (GPA: 3.52/4.00) :

Sichuan University of Science and Engineering, Zigong, China.

Main courses: Plant fiber chemistry, Principles and Engineering of Pulping and Mathematical Modeling.

(Sep 2018 - Sep 2021)

Chengdu Qingong Polytechnic University (GPA: 3.43/4.00) :

Formerly Chengdu Textile College, Chengdu, China.

Main courses: Textile Materials Science, Fabric structure and construction.

Research Experience

(Feb 2025 - Jan 2026) —

Multifunctional MXene/AgNW/Collagen Composite Yarn for Wearable Sensing and Biomedical Applications :

Supervisor: Prof. Guangming Cai and Dr. Chengfei Yue.

- Developed a hierarchical yarn structure combining conductive nanomaterials and bioactive components to achieve strain sensing and multiple biofunctional properties.
- Investigated the multifunctional performance of the composite yarn, including antibacterial activity, photothermal conversion, sustained drug release, and hemostatic properties for potential biomedical applications.
- Conducted material characterization and performance evaluation using techniques such as SEM, FTIR, mechanical testing, and antibacterial assays.

Status: Manuscript submitted to Science China Materials.

(May 2024 - Dec 2024) —

Tannic Acid-Mediated Collagen/Cellulose Nanofiber Composite Aerogels:

Supervisor: Prof. Guangming Cai and Dr. Chengfei Yue.

- Developed tannic-acid-mediated collagen self-assembled composite aerogels.
- Investigated the multifunctional biological performance of the aerogel, including antibacterial activity and rapid hemostatic capability for potential wound healing applications.
- Conducted material characterization and structural analysis using techniques such as SEM, FTIR, and mechanical performance testing.

Status: Published in ACS Applied Polymer Materials.

(Sep 2023 - May 2024) —

Ultrasound-Assisted Extraction of Collagen and Wet Spinning of Collagen Fibers :

Supervisor: Dr. Chengfei Yue.

- Used an ultrasound-assisted enzymatic extraction method to isolate collagen from porcine skin and optimize the extraction conditions.
- Fabricated collagen fibers via wet spinning and optimized spinning parameters to obtain continuous and uniform fibers.

Status: Published in Journal of Textile Engineering (in Chinese).

Honors and Awards



Languages

- Chinese: Native
- English: Intermediate - strong academic reading and writing skills, actively improving listening and speaking skills.
- English Test: Passed College English Test Band 4 (CET-4), China.

Technical Skills

- Materials Fabrication: Electrospinning, Aerogel fabrication, Collagen extraction, Wet spinning, friction spinning, Preparation of MXene.
- Characterization Techniques: SEM, FTIR, XRD, XPS, Mechanical testing.
- Experiments: Sensor performance evaluation, Antibacterial activity evaluation, In vitro hemostatic evaluation.
- Data Analysis and Software: Origin, Microsoft Office, Visual Basic for Applications (VBA), Mathematical modeling